

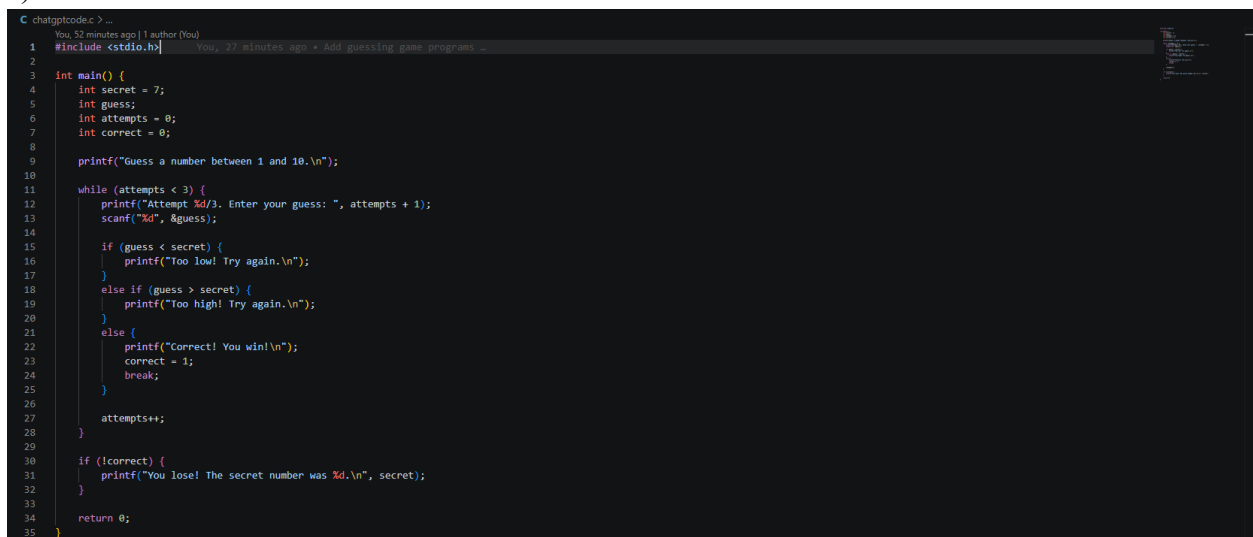
EECS 348 Software Engineering

a) I'm using ChatGPT and Claude for the two different AI programs.

b) I downloaded the Claude app from google, so I'm using the AI apps on my windows computer, and I'm using the google version of ChatGPT.

c) The prompt I'm using is a direct screenshot of the requirements for the assignment for the project from the assignment 1 instructions pdf. I want the program to get the most descriptive parameters, so the actual assignment outline is perfect as a prompt.

e)

A screenshot of a code editor window titled 'chatgptcode.c'. The code is a C program for a guessing game. It includes a header file <stdio.h> and defines a main function. Inside main, it initializes a secret number to 7, and variables for guess, attempts, and correct to 0. It prints a prompt to guess a number between 1 and 10. A while loop runs as long as attempts are less than 3. Inside the loop, it prints the attempt number and asks for a guess. It then checks if the guess is too low, too high, or correct. If correct, it increments the correct counter and breaks the loop. If not correct, it increments the attempts counter. After the loop, it checks if the user is correct and prints the secret number if they are not. Finally, it returns 0.

```
1 #include <stdio.h>
2
3 int main() {
4     int secret = 7;
5     int guess;
6     int attempts = 0;
7     int correct = 0;
8
9     printf("Guess a number between 1 and 10.\n");
10
11     while (attempts < 3) {
12         printf("Attempt %d/3. Enter your guess: ", attempts + 1);
13         scanf("%d", &guess);
14
15         if (guess < secret) {
16             printf("Too low! Try again.\n");
17         }
18         else if (guess > secret) {
19             printf("Too high! Try again.\n");
20         }
21         else {
22             printf("Correct! You win!\n");
23             correct = 1;
24             break;
25         }
26         attempts++;
27     }
28
29     if (!correct) {
30         printf("You lose! The secret number was %d.\n", secret);
31     }
32
33     return 0;
34 }
35
```

ChatGPT ^

```

1  #include <stdio.h>
2
3  int main() {
4      int secret = 7; // fixed secret number
5      int guess;
6      int attempts = 3;
7
8      printf("Guess a number between 1 and 10.\n");
9
10     int i;
11     for (i = 1; i <= attempts; i++) {
12         printf("Attempt %d/%d. Enter your guess: ", i, attempts);
13         scanf("%d", &guess);
14
15         if (guess == secret) {
16             printf("Correct! You win!\n");
17             break;
18         } else if (guess > secret) {
19             printf("Too high! Try again.\n");
20         } else {
21             printf("Too low! Try again.\n");
22         }
23     }
24
25     if (i > attempts) {
26         printf("Sorry, you lose! The number was %d.\n", secret);
27     }
28
29     return 0;
30 }

```

Claude[^]

e) Claude Analysis – For correctness Claude passes which was expected for such a simple program it should have created exactly what was asked for in the prompt. It uses the required parts like an if else statement and a while or for loop, while also handling too many guesses.

Execution Time – The time for the Claude code is $O(1)$ when you get it on the first try but can max out at $O(3)$ if you get it right on the last guess or don't get it at all. It's because of the three-limit loop which causes it to stack up with more guesses.

Space Complexity – The space complexity is also $O(1)$ since it's such a small program and only uses 3 variables (integers)

Maintainability – The code is very simple and just needs a good understanding of how a loop works and that we are increasing i and that as long as $i \leq \text{attempts}$ then the loop continues. Changes to the code would be minimal or even none since it can't be improved too much or added on.

ChatGPT Analysis – Correctness – The ChatGPT code is also perfect and correctly created the code with all the same requirements as the Claude code did. It uses a correct variable that will check on input if the guess and correct and then break the loop and win.

Execution Time – It has the same run time as the Claude code which is $O(1)$ with a maximum of rising to $O(3)$ if all guesses are used, or the answer isn't guessed.

Space Complexity – Also $O(1)$ while using 4 variables being the bigger storage and memory wise, but it's going to be so small it doesn't matter.

Maintainability – It also has very little and easy maintainability which is arguably easier to understand because it's based on the correct variable check. It has nothing to do with the loop and is a very simple straightforward check.

f) My preferred AI code would be the Claude one since to me, I'm comfortable with the for-loop concept and think it's easier to follow and more concise than creating an extra variable to check for the right answer.

g) To improve the Claude code, you can do a few things that were kind of overlooked since the code is meant to be simple and have perfect inputs. So first you'd change some of the input code by making the scanf check for an integer by running through an if statement and making sure it's not $\neq 1$. Then wipe the assignment with getchar() $\neq$ '\n' if a string or something else is assigned. Also, adding the correct check from the ChatGPT code is a good idea since it is technically simpler and better than using the loop to do the work. This doesn't change the time or space complexity, besides maybe a small negligible increase in memory. For maintainability, making a named constant to test it against also helps to make it simpler and clearer than what we're looping to. Add comments as well to help editors or debuggers identify code.